HRMS

10 訓練管理 — 系統設計書

HR10 TRAINING Service — System Design

系統名稱	版本
人力資源暨專案管理系統	**1.0**
日期	作者
2026-03-17	**HRMS 開發團隊**

目錄

訓練管理服務系統設計書

版本: 1.0 **日期:** 2025-12-07 **Domain代號:** 10 (TRN) **導入階段:** 第三階段（進階人資功能)

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1. 服務概述

1.1 核心功能

- ✅ **訓練計畫管理:** 年度計畫、預算控管
- ✅ **課程管理:** 內訓/外訓、線上/線下
- ✅ **報名審核:** 課程報名與審核
- ✅ **訓練時數追蹤:** 法定時數合規
- ✅ **證照管理:** 證照登錄、到期提醒

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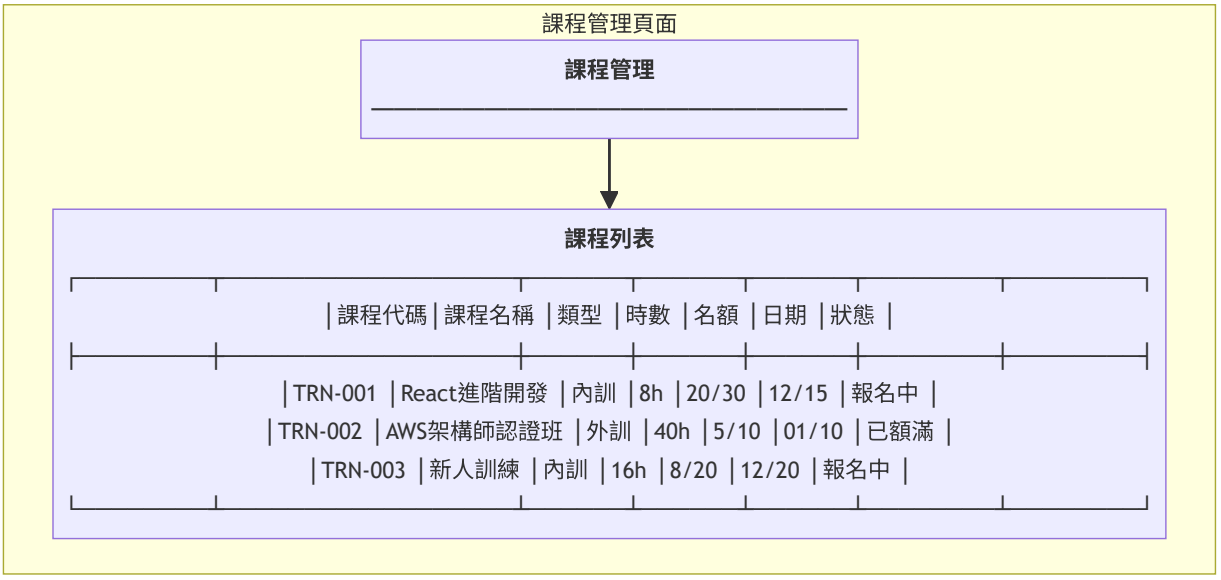
2. UI設計

2.1 頁面清單

頁面代碼	頁面名稱	路由
`HR10-P01`	課程管理頁面	`/admin/training/courses`
`HR10-P02`	課程報名頁面	`/admin/training/enrollments`
`HR10-P03`	我的訓練 (ESS)	`/profile/training`
`HR10-P04`	證照管理頁面	`/admin/training/certificates`
`HR10-P05`	我的證照 (ESS)	`/profile/certificates`
`HR10-P06`	訓練時數統計	`/admin/training/reports`

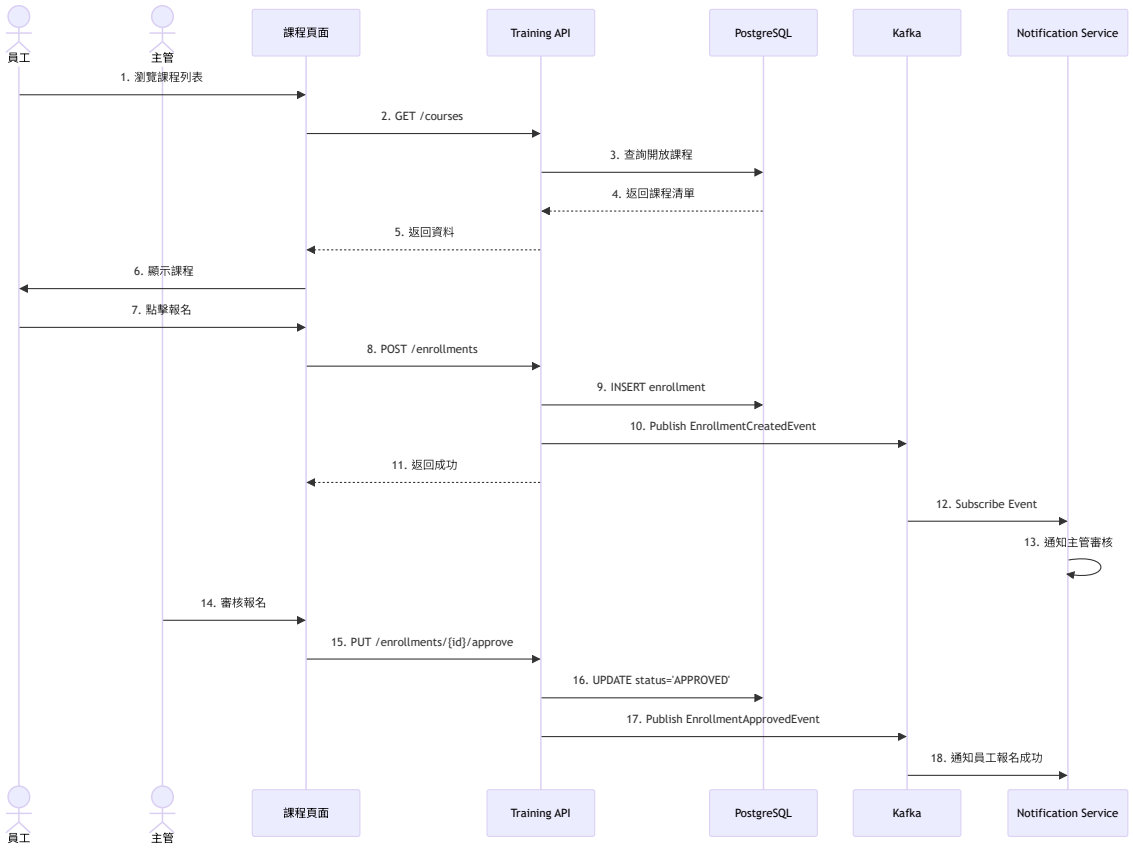
2.2 UI線稿

2.2.1 課程管理頁面 (HR10-P01)

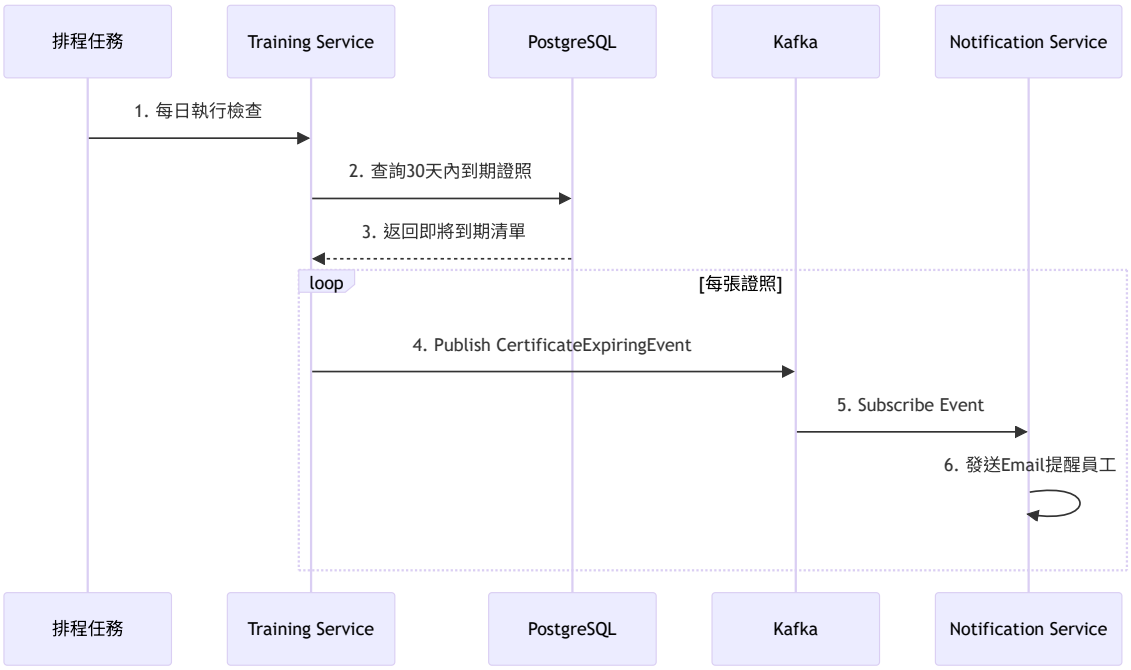


3. UX流程設計

3.1 課程報名流程



3.2 證照到期提醒流程



4. 畫面事件說明

4.1 課程管理頁面事件 (HR10-P01)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-CRS-01`	新增課程按鈕	onClick	開啟新增課程對話框	-
`E-CRS-02`	編輯按鈕	onClick	開啟編輯對話框	GET /api/v1/training/courses/{id}
`E-CRS-03`	關閉課程按鈕	onClick	關閉課程報名	PUT /api/v1/training/courses/{id}/close
`E-CRS-04`	新增課程確認	onClick	建立課程	POST /api/v1/training/courses

4.2 課程報名頁面事件 (HR10-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-ENR-01`	報名按鈕	onClick	建立報名	POST /api/v1/training/enrollments
`E-ENR-02`	審核按鈕	onClick	審核報名	PUT /api/v1/training/enrollments/{id}/approve
`E-ENR-03`	拒絕按鈕	onClick	拒絕報名	PUT /api/v1/training/enrollments/{id}/reject
`E-ENR-04`	取消報名按鈕	onClick	取消報名	PUT /api/v1/training/enrollments/{id}/cancel

4.3 我的證照頁面事件 (HR10-P05)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-CERT-01`	新增證照按鈕	onClick	開啟新增證照對話框	-
`E-CERT-02`	上傳附件	onChange	上傳證照檔案	POST /api/v1/training/certificates/upload
`E-CERT-03`	新增證照確認	onClick	建立證照記錄	POST /api/v1/training/certificates

5. Data Flow設計

5.1 前端狀態管理 (Redux)

5.1.1 State結構

```
interface TrainingState {
  // 課程管理
  courses: {
    list: TrainingCourse[];
    currentCourse: TrainingCourse | null;
    filters: {
      type: CourseType;
      status: CourseStatus;
    };
    loading: boolean;
  };

  // 報名管理
  enrollments: {
    list: Enrollment[];
    myEnrollments: Enrollment[];
    pendingApprovals: Enrollment[];
    loading: boolean;
  };

  // 證照管理
  certificates: {
    list: Certificate[];
    expiringCertificates: Certificate[];
    loading: boolean;
  };

  // 訓練時數統計
  statistics: {
    myHours: TrainingHours;
    requiredHours: number;
    loading: boolean;
  };
}
```

5.1.2 Redux Actions

```
export const courseActions = {
  fetchCourses: createAsyncThunk('training/fetchCourses', async () => {
    const response = await trainingService.getCourses();
    return response;
  }),

  createCourse: createAsyncThunk('training/createCourse', async (data: CreateCourseRequest)
    const response = await trainingService.createCourse(data);
    return response;
  }),
};

export const enrollmentActions = {
  enrollCourse: createAsyncThunk('training/enroll', async (courseId: string) => {
    await trainingService.enroll(courseId);
    return courseId;
  }),

  approveEnrollment: createAsyncThunk('training/approveEnrollment', async (enrollmentId: st
    await trainingService.approveEnrollment(enrollmentId);
    return enrollmentId;
  }),

  completeTraining: createAsyncThunk('training/complete', async ({enrollmentId, hours}: {en
    await trainingService.completeTraining(enrollmentId, hours);
    return { enrollmentId, hours };
  }),
};

export const certificateActions = {
  addCertificate: createAsyncThunk('training/addCertificate', async (data: AddCertificateRe
    const response = await trainingService.addCertificate(data);
    return response;
  }),

  fetchExpiringCertificates: createAsyncThunk('training/fetchExpiringCertificates', async (
    const response = await trainingService.getExpiringCertificates();
    return response;
  }),
};
```

6. 資料庫設計

```sql
-- 課程表
CREATE TABLE training_courses (
    course_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    course_code VARCHAR(50) NOT NULL UNIQUE,
    course_name VARCHAR(255) NOT NULL,
    course_type VARCHAR(20) CHECK (course_type IN ('INTERNAL', 'EXTERNAL')),
    delivery_mode VARCHAR(20) CHECK (delivery_mode IN ('ONLINE', 'OFFLINE', 'HYBRID')),
    instructor VARCHAR(100),
    duration_hours DECIMAL(5,1) NOT NULL,
    max_participants INTEGER,
    start_date DATE,
    end_date DATE,
    location VARCHAR(255),
    cost DECIMAL(10,2) DEFAULT 0,
    status VARCHAR(20) DEFAULT 'DRAFT' CHECK (status IN ('DRAFT', 'OPEN', 'CLOSED', 'COMPLE
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 報名表
CREATE TABLE training_enrollments (
    enrollment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    course_id UUID NOT NULL REFERENCES training_courses(course_id),
    employee_id UUID NOT NULL,
    status VARCHAR(20) DEFAULT 'REGISTERED' CHECK (status IN ('REGISTERED', 'APPROVED', 'AT
    attendance BOOLEAN DEFAULT FALSE,
    completed_hours DECIMAL(5,1),
    score DECIMAL(5,2),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_enrollment UNIQUE (course_id, employee_id)
);

-- 證照表
CREATE TABLE certificates (
    certificate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    certificate_name VARCHAR(255) NOT NULL,
    issuing_organization VARCHAR(255),
    certificate_number VARCHAR(100),
    issue_date DATE NOT NULL,
    expiry_date DATE,
    attachment_url VARCHAR(500),
    is_verified BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_cert_employee ON certificates(employee_id);
CREATE INDEX idx_cert_expiry ON certificates(expiry_date);
```

4. Domain設計

```java
@Entity
public class TrainingEnrollment {
    @EmbeddedId
    private EnrollmentId id;
    private UUID courseId;
    private UUID employeeId;

    @Enumerated(EnumType.STRING)
    private EnrollmentStatus status;

    private boolean attendance;
    private BigDecimal completedHours;

    /**
     * 確認出席
     */
    public void confirmAttendance(BigDecimal hours) {
        this.attendance = true;
        this.completedHours = hours;
        this.status = EnrollmentStatus.ATTENDED;
    }

    /**
     * 完成訓練
     */
    public void complete() {
        if (!this.attendance) {
            throw new DomainException("未出席無法完成");
        }
        this.status = EnrollmentStatus.COMPLETED;

        DomainEventPublisher.publish(new TrainingCompletedEvent(
            this.employeeId, this.courseId, this.completedHours
        ));
    }
}
```

8. 領域事件設計

8.1 事件清單

事件名稱	觸發時機	訂閱服務	業務影響
`EnrollmentCreatedEvent`	員工報名課程	Notification	通知主管審核
`EnrollmentApprovedEvent`	主管審核通過	Notification	通知員工報名成功
`TrainingCompletedEvent`	員工完成訓練	Reporting	更新訓練時數統計
`CertificateExpiringEvent`	證照即將到期	Notification	提醒員工更新證照

8.2 事件Schema

8.2.1 EnrollmentCreatedEvent

```json
{
  "eventId": "uuid",
  "eventType": "EnrollmentCreated",
  "aggregateId": "enrollment-id",
  "aggregateType": "TrainingEnrollment",
  "occurredAt": "2025-12-07T10:00:00Z",
  "payload": {
    "enrollmentId": "uuid",
    "courseId": "course-uuid",
    "courseName": "React進階開發",
    "employeeId": "emp-uuid",
    "employeeName": "張三",
    "managerId": "mgr-uuid",
    "managerName": "李組長",
    "trainingHours": 8,
    "cost": 5000,
    "reason": "提升前端技術能力"
  },
  "metadata": {
    "userId": "emp-uuid",
    "userName": "張三",
    "source": "Training Service",
    "version": "1.0"
  }
}
```

8.2.2 TrainingCompletedEvent

```
{
  "eventId": "uuid",
  "eventType": "TrainingCompleted",
  "aggregateId": "enrollment-id",
  "aggregateType": "TrainingEnrollment",
  "occurredAt": "2025-12-15T17:00:00Z",
  "payload": {
    "enrollmentId": "uuid",
    "employeeId": "emp-uuid",
    "employeeName": "張三",
    "courseId": "course-uuid",
    "courseName": "React進階開發",
    "completedHours": 8,
    "completedDate": "2025-12-15",
    "score": 85,
    "passed": true
  },
  "metadata": {
    "userId": "trainer-uuid",
    "userName": "講師",
    "source": "Training Service",
    "version": "1.0"
  }
}
```

8.2.3 CertificateExpiringEvent

```
{
  "eventId": "uuid",
  "eventType": "CertificateExpiring",
  "aggregateId": "certificate-id",
  "aggregateType": "Certificate",
  "occurredAt": "2025-12-07T08:00:00Z",
  "payload": {
    "certificateId": "uuid",
    "employeeId": "emp-uuid",
    "employeeName": "張三",
    "employeeEmail": "zhang@company.com",
    "certificateName": "AWS架構師證照",
    "issuingOrganization": "Amazon",
    "issueDate": "2022-12-15",
    "expiryDate": "2026-01-15",
    "daysUntilExpiry": 30,
    "isRequired": true
  },
  "metadata": {
    "userId": "system",
    "userName": "System Scheduler",
    "source": "Training Service",
    "version": "1.0"
  }
}
```

---

9. API設計 (10個端點)

端點	方法	Controller
`/api/v1/training/courses`	POST	HR10CourseCmdController
`/api/v1/training/courses`	GET	HR10CourseQryController
`/api/v1/training/enrollments`	POST	HR10EnrollmentCmdController
`/api/v1/training/enrollments/{id}/approve`	PUT	HR10EnrollmentCmdController
`/api/v1/training/enrollments/{id}/complete`	PUT	HR10EnrollmentCmdController
`/api/v1/training/my`	GET	HR10EnrollmentQryController
`/api/v1/training/certificates`	POST	HR10CertificateCmdController
`/api/v1/training/certificates`	GET	HR10CertificateQryController
`/api/v1/training/certificates/expiring`	GET	HR10CertificateQryController
`/api/v1/training/my/hours`	GET	HR10ReportQryController

10. 工項清單摘要

10.1 前端開發工項 (總計: 64小時)

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
F-TRN-01	**課程管理頁面 (HR10-P01)**	**頁面開發**	**12**	**features/training**	
F-TRN-01-01	CourseListComponent	元件	3	components/	課程列表表格
F-TRN-01-02	CreateCourseModal	元件	4	components/	新增課程對話框
F-TRN-01-03	useCourseManagement	Hook	3	hooks/	課程管理邏輯
F-TRN-01-04	單元測試	測試	2	tests/	元件測試
F-TRN-02	**課程報名頁面 (HR10-P02)**	**頁面開發**	**14**	**features/training**	
F-TRN-02-01	EnrollmentListComponent	元件	4	components/	報名列表
F-TRN-02-02	ApprovalModal	元件	4	components/	審核對話框
F-TRN-02-03	useEnrollment	Hook	3	hooks/	報名邏輯
F-TRN-02-04	單元測試	測試	3	tests/	Hook測試
F-TRN-03	**我的訓練頁面 (HR10-P03)**	**頁面開發**	**10**	**features/training**	
F-TRN-03-01	MyTrainingComponent	元件	4	components/	我的訓練頁面
F-TRN-03-02	TrainingHistoryList	元件	3	components/	訓練歷程列表
F-TRN-03-03	單元測試	測試	3	tests/	元件測試
F-TRN-04	**證照管理頁面 (HR10-P04/P05)**	**頁面開發**	**14**	**features/training**	
F-TRN-04-01	CertificateListComponent	元件	4	components/	證照列表

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
F-TRN-04-02	AddCertificateModal	元件	4	components/	新增證照對話框
F-TRN-04-03	ExpiringAlert	元件	2	components/	到期提醒元件
F-TRN-04-04	useCertificate	Hook	2	hooks/	證照管理邏輯
F-TRN-04-05	單元測試	測試	2	tests/	元件測試
F-TRN-05	**訓練時數統計頁面 (HR10-P06)**	**頁面開發**	**10**	**features/training**	
F-TRN-05-01	TrainingStatistics	元件	4	components/	時數統計頁面
F-TRN-05-02	HoursChart	元件	3	components/	時數圖表
F-TRN-05-03	單元測試	測試	3	tests/	圖表測試
F-TRN-06	**API & Redux 整合**	**整合**	**4**	**features/training**	
F-TRN-06-01	TrainingApi	API	2	api/	API呼叫封裝
F-TRN-06-02	trainingSlice	Redux	2	store/	State管理

前端小計: 64小時

10.2 後端開發工項 (總計: 56小時)

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
B-TRN-01	**Domain層開發**	**領域模型**	**18**	**domain**	
B-TRN-01-01	TrainingCourse聚合根	聚合根	4	domain/model	課程管理
B-TRN-01-02	TrainingEnrollment聚合根	聚合根	5	domain/model	報名管理(含狀態)
B-TRN-01-03	Certificate聚合根	聚合根	3	domain/model	證照管理
B-TRN-01-04	領域事件定義	事件	3	domain/event	4個領域事件
B-TRN-01-05	Domain單元測試	測試	3	domain/	100%覆蓋率
B-TRN-02	**Repository層開發**	**持久化**	**12**	**infrastructure**	
B-TRN-02-01	ITrainingCourseRepository	介面	1	domain/repository	Repository介面
B-TRN-02-02	TrainingCourseRepositoryImpl	實作	3	infrastructure/repository	MyBatis實作
B-TRN-02-03	IEnrollmentRepository	介面	1	domain/repository	Repository介面
B-TRN-02-04	EnrollmentRepositoryImpl	實作	3	infrastructure/repository	含審核查詢
B-TRN-02-05	ICertificateRepository	介面	1	domain/repository	Repository介面
B-TRN-02-06	CertificateRepositoryImpl	實作	2	infrastructure/repository	含到期查詢
B-TRN-02-07	Repository測試	測試	1	infrastructure/	整合測試
B-TRN-03	**Application層開發**	**應用服務**	**16**	**application**	
B-TRN-03-01	CreateCourseServiceImpl	Command	2	application/service	建立課程
B-TRN-03-02	EnrollCourseServiceImpl	Command	2	application/service	報名課程

工項ID	工項名稱	類別	工時(小時)	負責模組	說明
B-TRN-03-03	ApproveEnrollmentServiceImpl	Command	2	application/service	審核報名
B-TRN-03-04	CompleteTrainingServiceImpl	Command	2	application/service	完成訓練(發事件)
B-TRN-03-05	AddCertificateServiceImpl	Command	2	application/service	新增證照
B-TRN-03-06	GetMyTrainingsServiceImpl	Query	2	application/service	查詢我的訓練
B-TRN-03-07	GetExpiringCertificatesServiceImpl	Query	2	application/service	查詢到期證照
B-TRN-03-08	Service測試	測試	2	application/	服務測試
B-TRN-04	**Interface層開發**	**API**	**8**	**interface**	
B-TRN-04-01	HR10CourseCmdController	Controller	2	interface/api	課程Command
B-TRN-04-02	HR10EnrollmentCmdController	Controller	2	interface/api	報名Command
B-TRN-04-03	HR10CertificateCmdController	Controller	1	interface/api	證照Command
B-TRN-04-04	HR10*QryController	Controller	2	interface/api	Query Controllers
B-TRN-04-05	Request/Response DTO	DTO	1	interface/api	10個端點DTO
B-TRN-05	**排程任務開發**	**排程**	**2**	**infrastructure**	
B-TRN-05-01	CertificateExpiryScheduler	排程	2	infrastructure/scheduler	每日檢查到期證照

後端小計: 56小時

10.3 工項總覽

類別	前端工時	後端工時	小計
頁面開發	60	-	60
API整合	4	-	4
Domain層	-	18	18
Repository層	-	12	12
Application層	-	16	16
Interface層	-	8	8
排程任務	-	2	2
總計	**64**	**56**	**120**

10.4 人力配置建議

- **前端工程師:** 1人，預計1.5週完成 (64小時)
- **後端工程師:** 1人，預計1.5週完成 (56小時)
- **總開發週期:** 2週 (含測試與整合)

10.5 開發優先順序

1. **Phase 1 (Week 1):**
    - 後端: Domain層 + Repository層 + 基礎API (30小時)
    - 前端: 課程管理 + 課程報名 (26小時)
2. **Phase 2 (Week 2):**
    - 後端: Application層 + Interface層 + 排程任務 (26小時)
    - 前端: 證照管理 + 訓練時數統計 + 整合測試 (38小時)

10.6 技術挑戰與風險

挑戰項目	風險等級	解決方案
證照到期排程	低	使用Spring Scheduler
訓練時數統計	中	考慮使用Read Model (CQRS)
外訓課程同步	低	手動匯入，未來考慮API整合

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